M VIJAY SUMANTH

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PROJECTS

BharatCare — AI-Powered Regional Healthcare Assistant (Live Demo | GitHub) Sep 2025 – Feb 2026

React, TypeScript, Flask, Scikit-learn, Groq LLaMA

- Led a team to build a full-stack health-guidance web app predicting likely diseases from user-reported symptoms using a Random Forest classifier trained on a curated dataset of 518 symptom-disease mappings (React/TypeScript, Flask REST API)
- Designed a rule-based environmental risk engine combining real-time weather, AQI, season, and UV index data (OpenWeather API) with ML predictions to flag context-specific health risks
- Integrated Groq LLaMA 3.1 for plain-language AI health explanations and added multilingual support across 10 Indian languages via LibreTranslate, with Firebase-based auth and prediction history

Published as: "BharatCare: An AI-Powered Regional Healthcare Assistant using Machine Learning and Environmental Risk Analysis," IRJET, Vol. 13, Issue 3, March 2026

Crop Disease Detection System (GitHub) Jan 2025 – Apr 2025

Python, TensorFlow/Keras, Streamlit

- Led a team to train a CNN-based image classifier to detect 15 crop disease classes across pepper, potato, and tomato plants, using ~16,000 images from the PlantVillage dataset
- Achieved 83.84% validation accuracy (weighted F1: 0.84) on a held-out validation set of 4,530 images, using EarlyStopping and checkpointing to prevent overfitting
- Built and deployed a Streamlit web app for real-time disease prediction from uploaded leaf images

Secure Password Manager (Live Demo | GitHub) Sep 2024 – Mar 2025

Node.js, Express, MySQL

- Built a full-stack password manager with JWT-based authentication and bcrypt password hashing
- Implemented authenticated AES-256-GCM encryption for stored passwords, including a migration path that upgrades legacy AES-256-CBC records to GCM on update without breaking existing data
- Designed full CRUD functionality plus a password history feature, a secure password generator, and a strength checker, with protected client-side routing

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, HTML, CSS

Frameworks/Libraries: React, Flask, Node.js, Express, Tailwind CSS

Databases: MySQL, MongoDB, Firebase

Tools/Platforms: Git, GitHub, VS Code, Render, Vite, AWS (S3 static hosting)

Other: REST API Integration, JWT & bcrypt Authentication, LLM API Integration (Groq)

CERTIFICATIONS

- MERN Full Stack Development — Apna College
- Web Development + DevOps + Blockchain Cohort — 100xDevs, Cohort 3
- McKinsey Forward Program — McKinsey.org
- Introduction to Generative AI — Google Cloud, via Simplilearn SkillUp

EDUCATION

Joginpally B.R. Engineering College, Hyderabad Nov 2022 – Jun 2026

B.Tech, Computer Science Engineering — CGPA: 7.47/10

Kendriya Vidyalaya No.1 Golconda, Hyderabad Apr 2010 – May 2022

Schooling (10th & 12th)